

Shahriar Siam

Department of Applied Chemistry and Chemical Engineering, UNIVERSITY OF RAJSHAHI, Rajshahi 6205, Bangladesh

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 [LinkedIn](#)

Education

University of Rajshahi B.Sc. Engineering Program in Applied Chemistry and Chemical Engineering 11/2022 - Present

- Expected Graduation: January 2027

Jalalabad Cantonment Public School & College, Sylhet, Bangladesh

06/2019 - 12/2021

- Completed Higher Secondary Level
- GPA: 5.00/5.00

Skills

- **Software skills:** Python (Beginner), Django (Intermediate), MS Word (Proficient), MS Excel (Proficient), MS PowerPoint (Proficient), Google Forms (Proficient), Gaussian 16 (Proficient), DFT (Proficient), Unisim (Intermediate),
- **Technical skills:** Machine Learning (Beginner), Web Development with Django, Basic knowledge of industrial instruments and processes
- **Language skills:** Written English (proficient), Oral English (intermediate), Bengali (proficient)

RESEARCH EXPERIENCE

Substituent Effects on O–H BDE in Phenolic Antioxidants: A DFT Study

Supervisor: **Dr. Jewel Hossen**, Department of Chemistry, RUET

- Performed DFT calculations using Gaussian 16 at the B3LYP/6-311+G(d,p) level to investigate substituent effects ($-\text{CH}_3$, $-\text{NH}_2$, $-\text{OCH}_3$, $-\text{SH}$) on O–H bond dissociation energy (BDE) in phenol derivatives
- Conducted BDE, HOMO-LUMO gap, MEP surface, NBO charge, and DOS analyses to evaluate antioxidant activity
- Results presented at two international conferences; manuscript under review in Discover Chemistry

Advances in Perovskite Solar Cells: Construction and Efficiency

Supervisor: **Prof. Md. Saiful Islam**, Department of Applied Chemistry & Chemical Engineering, RU

- Contributed as co-author to a comprehensive review on perovskite solar cell design and efficiency optimization
- Manuscript under review in Solar Energy (Elsevier)

PUBLICATION

Journal Articles

1. Jewel Hossen, Md Forhad Babu, Tanu Dey, **Shahriar Siam**, Md Mazharul Islam, Md Masum Iqbal, Apel Mahamud Rabby, and Md Tanvir Rashid Rezvi, “Computational Study of the Effects of Substituent Type and Position on the O–H Bond Dissociation Energy of Phenolic Compounds,” Discover Chemistry, 2026. DOI: <https://doi.org/10.1007/s44371-026-00856-6>

2. Md Saiful Islam, **Shahriar Siam**, Md Forhad Babu, Tanu Dey, Md Mazharul Islam, Sraboni Akter, Jahid Hasan Tareq, and Promita Poroma Sarker, “Advances in Perovskite Solar Cells: Construction and Efficiency,” Solar Energy, Elsevier, 2026. **(Under Review)**
3. Md Nabil Bin Zakir, Md Toukir Biswas, **Shahriar Siam**, and Raghieb Mohammad Altafin, “Toward Sustainable Alternatives: Analyzing Bioplastic Supply Chain Challenges in Bangladesh’s Local Market,” Supply Chain Insider, ISSN: 2617-7420, 2025.
<https://supplychaininsider.org/ojs/index.php/home/article/view/133>

Conference Proceedings

1. Jewel Hossen, **Shahriar Siam**, Md Forhad Babu, Tanu Dey, Md Mazharul Islam, Md Masum Iqbal, Apel Mahamud Rabby, and Md Tanvir Rashid Rezvi, “Substituent Effects on O–H Bond Dissociation Energy in Phenolic Antioxidants: A DFT Approach,” ICEI-2026, Abstract ID: TCS05-238, May 7–8, 2026, Atomic Energy Centre, Dhaka. **(Oral)**
2. Jewel Hossen, Md Forhad Babu, Tanu Dey, **Shahriar Siam**, Md Mazharul Islam, Md Masum Iqbal, Apel Mahamud Rabby, and Md Tanvir Rashid Rezvi, “Rational Design of Phenolic Antioxidants: A DFT Study on Substituent-Driven Modulation of O–H Bond Dissociation Energies,” 6th Young Scientist Congress, Abstract ID: PP059, April 4–5, 2026, Dhaka. **(Poster)**
3. Md Saiful Islam, Juel Mia, Md Forhad Babu, **Shahriar Siam**, Md Mazharul Islam, et al., “Green Corrosion Inhibition of Steels Using Moringa oleifera Leaves Extract in Acidic Medium,” ICESGE-2025, Abstract ID: PP_ICESGE_036, December 13, 2025, University of Dhaka. **(Poster)**
4. Hossain Md. Al-Amin, Meer Tozammel Hossain, Md Ehsanul Haque Chowdhury, Maidha Hussain, Mashrafi Sultana Shimu, and **Shahriar Siam**, “Exploring PTSD in the Aftermath of the 2024 Quota Movement: Demographic and Psychological Insights from Bangladesh,” 4th International Scientific Research & Innovation Congress, December 14–16, 2025, Bursa, Türkiye. **(Oral)**
5. Anik Chandro Paul, Nasrin Akter Tithe, Sourov Pal, Mohammad Taufiq Alam, **Shahriar Siam**, Tanu Dey, Md Forhad Babu, and Md Mazharul Islam, “Computational Screening of Hydnocarpus kurzii (Chaulmoogra) Compounds as HER2 Inhibitors for Breast Cancer Therapy,” ICLESSD-2025, Abstract ID: BioP10, December 18–19, 2025, Jagannath University, Dhaka.

ORGANIZATIONAL EXPERIENCE

Organisation: POSITIVE PLUS FOUNDATION

- **Position:** National Officer of Quality Education
- **Role:** Assisting in planning and implementing education-based programs aligned with SDG 4.

Contributed to awareness campaigns, virtual education sessions, and reporting efforts for educational impact.

- **Duration:** 22/09/2024 – Present

Organisation: Rajshahi University Research Society

- **Position:** Joint General Secretary
- **Role:** Supporting the overall management and administration of the organization, coordinating organizational activities and team members, assisting in planning and executing research-related initiatives and events, maintaining effective communication and documentation, and ensuring smooth coordination among different wings and teams.
- **Duration:** July 2026 - present

Organisation: Rajshahi University Research Society

- **Position:** IT Director

- **Role:** Leading and managing IT operations of the organization, maintaining digital platforms and tools, supporting online research activities and events, ensuring smooth technical workflows, and coordinating with team members for system-based reporting and documentation.

- **Duration:** April 2025 – July 2026

Organisation: Materials Research Society Student Chapter, University of Dhaka

- **Position:** Campus Ambassador (University of Rajshahi)

- **Role:** Promoted materials science research initiatives, engaged students in academic and research activities, and served as a liaison between the chapter and the University of Rajshahi student community.

- **Duration:** 2025

Organisation: Rajshahi University Research Society (RURS)

- **Position:** Co-Lead, IT Team

- **Role:** Co-leading the IT team for the 1st International Research Conference and Research Fair, managing digital platforms and technical workflows, supporting virtual and on-site conference activities, coordinating technical support, and ensuring smooth execution of IT-related operations and reporting.

- **Duration:** 2024 – 2025

AWARDS AND ACHIEVEMENTS

- Inter-Cantonment Math Olympiad: 8th Place (National Round), 2018
- Physics Olympiad: 7th Place (Divisional Round), 2021
- Best Co-Lead Award: 1st International Student Conference and Research Fair 2024 – RURS